

Homework 39: Singular Value Decomposition SVD

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BCS A, E, F, G

Problem 1. For the matrices given below, you are required to find the following:

1. Find the singular value decomposition.
2. Find the dyadic (or outer product) expansion, also known as rank 1 matrix decomposition. The expression is

$$\sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T.$$

Equivalent Statement. Write the given matrix as sum of rank 1 matrices ($M_i = \sigma_i \mathbf{u}_i \mathbf{v}_i^T$, $\text{rank}(M_i) = 1, \forall i$).

3. Find the Moore-Penrose inverse

$$A^+ = V \Sigma^+ U^T.$$

Equivalently, it can be written as

$$A^+ = \sum_{i=1}^r \frac{1}{\sigma_i} \mathbf{v}_i \mathbf{u}_i^T.$$

Note: You are required to use both of versions to find A^+ .

4. Use singular value decomposition to construct bases of the four fundamental subspaces.

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

$$B = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

$$C = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

$$D = \begin{bmatrix} 1/2 & 1/2 \\ -1 & 1 \\ -1/2 & -1/2 \end{bmatrix}.$$

Submission Date: November 17, 2024 (Sunday – 11:59PM)

Good Luck